

Computer vision exercise 6 (page 1)

Task 17.

a) Purpose:

- Both matrices are used to relate a point x in an image to the epipolar line in another image.

Differences:

- Essential matrix uses normalized coordinates while fundamental matrix uses pixel coordinates.
- Essential matrix gives us the relative rotation and ~~translation~~ translation between the cameras, or their extrinsic parameters.
- Fundamental matrix gives us, apart from R, t , ~~it gives~~ the intrinsic parameters.

b) we have:

$$F = K'^{-T} E K^{-1} \quad (\text{lecture 12 slides p24})$$

$$\Rightarrow E = K'^T F K$$

The additional info we need is:

- The calibration matrices of the 2 cameras.

c) Number of DOF the fundamental matrix has.

- F has 7 degrees of freedom (DOF).

Explanation:

- F is a 3×3 matrix $\Rightarrow$ has 9 DOF
 - F is defined up to a scale factor $\Rightarrow$ ~~has~~ 1 DOF removed.
 - F is rank 2 (singular) $\Rightarrow \det(F) = 0 \Rightarrow$ 1 DOF removed.
- $\Rightarrow F$ has 7 DOF.

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1d) E has 5 DOF.

Explanation:

- E has Rotation $\Rightarrow$ has 3 DOF.
 - E has translation $\Rightarrow$ 3 more DOF.
 - E is defined up to an unknown scale $\Rightarrow$ 1 DOF removed
- $\Rightarrow$ E has: $3+3-1=5$ DOF.

Cyan: manually localized points Red: projected points



Cyan: manually localized points Red: projected points



Estimated width & height:

Picture width: 139.74 mm

Picture height: 107.02 mm

